



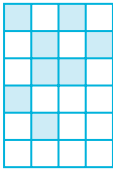
TEST YOURSELF 12

ANSWERS ON P. 607

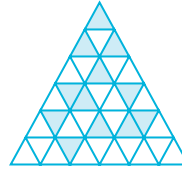
12.01

1 In each diagram, find the ratio of shaded parts to unshaded parts.

a



b



12.01

2 In a construction business, there are 17 women and 38 men. Find the ratio of:

a men to women

b women to the total

12.02

3 Copy and complete each pair of equivalent ratios.

a $2 : 3 = 8 : \underline{\hspace{1cm}}$

b $1 : 5 = 2 : \underline{\hspace{1cm}}$

c $2 : 3 : 4 = 6 : \underline{\hspace{1cm}} : \underline{\hspace{1cm}}$

12.02

4 Simplify each ratio.

a $12 : 21$

b $25 : 75$

c $6 : 36$

d $25 : 45$

e $18 : 6 : 24$

f $5 : 25 : 100$

g $2 \text{ cm} : 10 \text{ mm}$

h $5 \text{ dozen eggs} : 6 \text{ eggs}$

i $3 \text{ L} : 600 \text{ mL}$

j $50 \text{ m} : 1 \text{ km}$

k $20 \text{ minutes} : 1 \text{ hour}$

l $40 \text{ cents} : \$2$

m $18 \text{ months} : 3 \text{ years}$

n $12 \text{ days} : 3 \text{ weeks}$

o $4 \text{ days} : 10 \text{ hours}$

12.03

5 a The ratio of girls to boys in Year 7 is $4 : 3$. If there are 75 boys in Year 7, find how many girls there are.

b Tyrone and Erin invested in a business in the ratio $5 : 7$. If Erin invested \$63 000, how much did Tyrone invest?

12.04

6 Write each statement as a rate.

a 150 000 voters for 50 politicians

b 1000 sheep over 5 hectares

c 10 tickets for \$8

d 810 stitches to make 3 balls

12.05

7 For each pair, determine which is the better buy.

a Dog biscuits: 400 g of Woof for \$5.20 or 1 kg of Chow for \$11.10.

b Milk: 1 L of Dairy Farm for \$2.85 or 1250 mL of Milky for \$3.90.

12.06

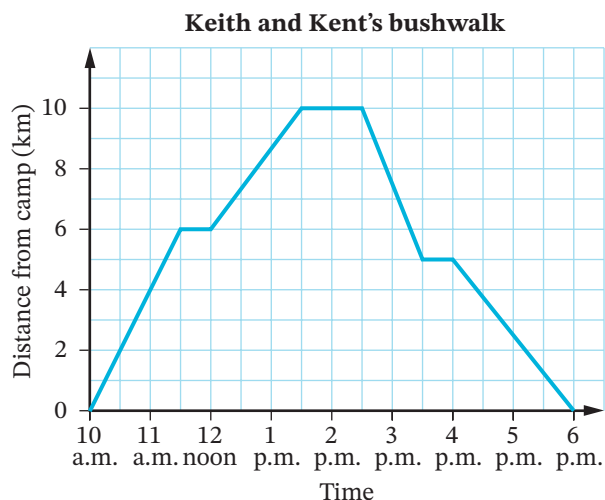
8 a Marie earns \$22.45 per hour. How much is she paid for 38 hours work?

b How many litres of petrol can you buy with \$65 if petrol costs \$1.79/L?
Answer correct to one decimal place.

c Fertiliser is used at 20 kg/ha. How many hectares can be covered with 150 kg of fertiliser?

- 9** Keith and Kent decided to go bushwalking one day. This travel graph shows their journey.

- How far did Keith and Kent walk?
- How many stops did they make?
- At what time did they start back?
- Between what times were Keith and Kent walking the fastest?
- Calculate their speed in the last 2 hours.



12.07

- 10** Round each time to the nearest hour.

- a** 9 h 50 min **b** 3.2 h **c** 4 h 12 min 49 s

12.08

- 11** Convert:

- 316 minutes to hours and minutes
- 208 seconds to minutes and seconds

12.08

- 12** Simplify each expression.

- $6 \text{ h } 45 \text{ min} + 3 \text{ h } 20 \text{ min}$
- $3 \text{ h } 16 \text{ min} + 1 \text{ h } 26 \text{ min}$
- $4 \text{ h } 33 \text{ min} - 2 \text{ h } 24 \text{ min}$
- $2 \text{ h } 19 \text{ min} - 1 \text{ h } 50 \text{ min}$

12.08

- 13** Convert to 12-hour time.

- a** 01:48 **b** 06:24 **c** 15:50 **d** 12:30

12.09

- 14** Convert to 24-hour time.

- a** 2:10 p.m. **b** 5:33 a.m. **c** 12:19 a.m. **d** 1:50 p.m.

12.09

- 15** Calculate the time difference between each pair of times.

- 5:25 a.m. and 9:45 a.m.
- 11:55 p.m. and 7:30 a.m.
- 1:15 p.m. and 8:10 p.m.
- 07:50 and 14:20
- 23:45 and 00:15
- 15:30 and 03:25

12.10

- 16** Samantha was born on 3rd June 2004. Calculate her age in years, months and days on 16th August 2023.

12.10

- 17** Refer to the train timetable on page 536.

- How long does the 7:45 p.m. train take to get to Sydney?
- When is the latest time you need to catch the train at Moss Vale to be in Sydney by 11 a.m.?

12.11